TASK 03

EVALUATION OF THE RELATIONSHIP BETWEEN A DRUG AND AN ADVERSE REACTION

Using Standard Causality Assessment Methods

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Evaluation of Drug–ADR Relationship

Standard causality assessment methods

Evaluation of the relationship between a drug and an adverse reaction using standard causality assessment methods.

Causality assessment

This task focuses on evaluating whether a suspected medicine is related to an adverse drug reaction (ADR), using the Naranjo Algorithm and the WHO-UMC causality approach.

Methods covered

01 Naranjo algorithm
02 WHO-UMC causality assessment

1. Aim

To evaluate the relationship between a suspected drug and an adverse drug reaction (ADR) using standard causality assessment methods and classify the reaction as Certain, Probable, or Possible.

2. Objectives

01	To understand the relationship between drug exposure and an adverse reaction.
02	To apply the Naranjo Algorithm to suspected ADR cases.
03	To assess cases using the WHO-UMC causality approach.
04	To classify suspected ADRs as Certain, Probable/Likely, or Possible.
05	To understand the importance of causality assessment in pharmacovigilance.

Standard Causality Assessment Methods

01 Naranjo algorithm

02 WHO-UMC causality assessment

Case Study 1

Amoxicillin and allergic reaction

Patient Details

Parameter	Information
Patient	Riya Rao
Age/Sex	28-year-old female
Weight	58 kg
Suspected drug	Amoxicillin
Dose	500 mg orally
Reaction	Red, itchy skin rash with mild facial swelling
Onset	After the 3rd dose
Dechallenge	Reaction improved after discontinuation
Rechallenge	Not performed
Alternative cause	No clear alternative cause identified

Clinical Assessment

The patient developed an itchy rash and facial swelling after repeated exposure to amoxicillin. The timing was compatible with a drug-related allergic reaction. The suspected medicine was discontinued, following which the symptoms improved.

Key observations

Timing: Reaction appeared after the 3rd dose

Dechallenge: Symptoms improved after discontinuation

Rechallenge: Not performed

Alternative cause: No clear alternative cause identified

Naranjo Algorithm Assessment

Questions, recorded answers and scores

Naranjo question	Answer	Score
Previous reports of this reaction?	Yes	+1
Reaction appeared after drug administration?	Yes	+2
Reaction improved after withdrawal?	Yes	+1
Reaction reappeared after re-exposure?	No / not done	0
Alternative causes present?	No	+2
Reaction with placebo?	Not done	0
Drug detected in toxic concentrations?	Not applicable	0

Naranjo Assessment — Continued

Naranjo question	Answer	Score
Dose-response relationship?	Not established	0
Previous similar reaction?	No information	0
Objective evidence	Yes	1

Result

Naranjo score = 7 → Probable

WHO-UMC classification: Probable/Likely

FINAL CLASSIFICATION	PROBABLE ADR

THANK

YOU

DRUG SAFETY • CAUSALITY ASSESSMENT